

ARK-447 Preregistration — Noise-Suppression Comparison: Dynamical Decoupling and Pauli Twirling

Experiment ID: ARK-447

Track: ExecutionProof authorization-boundary characterization

Status: PREREGISTRATION (not yet executed)

Protocol: Field 27 (LOCK → SPAM gate → principal job → analyze → verdict)

Target backends: `ibm_marrakesh` or `ibm_fez` (IBM Quantum Heron r2)

Date preregistered: 2026-07-17

Principal investigator: Derek Hone, Remnant Fieldworks Inc.

POST-HOC ERRATUM (added 2026-07-17, after execution) — do not remove

This document is preserved **as originally LOCKed** (tag `ark-447-v1.0-lock`, commit `c2466ff`) for provenance. Two items in this preregistration were found to be methodologically flawed and were **corrected in the executed analysis and in RESULTS.md (tag `ark-447-v1.1`)**. The original text below is left unchanged; this banner records the corrections:

1. **Invalid SPAM correction (§4.1).** The preregistered DENY criterion defined `L_D_corrected = max(0, L_D_raw - SPAM_P)`. This is **not a valid readout-error correction**. `SPAM_P` is the probability that a prepared $|+\rangle$ superposition reads '1' (≈ 0.5 by design — its expected physical behavior), not a spurious-excitation baseline. Subtracting ≈ 0.4944 from the raw DENY leakage (≈ 0.0013) forces every DENY result to zero and artificially inflates the boundary margin. **As executed, DENY leakage is reported RAW** (no subtraction); `SPAM_P` is retained only as a **gating diagnostic** confirming the payload readout axis behaves as expected. Both configurations pass the 0.02 DENY ceiling by enormous margins on raw leakage.
2. **Scope reduced to Pauli twirling vs. baseline.** Dynamical Decoupling was **omitted** during execution due to scheduling complexity, so the study is a Pauli-twirling-vs-baseline comparison (4 circuits), not a three-configuration DD/twirling survey. See `RESULTS.md`.

Corrected headline numbers: Baseline $S_A=0.9824$, $L_D_{\text{raw}}=0.0013$, $\Delta_B=0.7811$ (PASS); Pauli twirling $S_A=0.9875$, $L_D_{\text{raw}}=0.0012$, $\Delta_B=0.7863$ (PASS). ALLOW improvement $+0.0051$ is statistically significant (two-proportion $z=2.70$, $p=0.007$).

1. Central Question

Do noise-mitigation strategies improve authorization boundary fidelity on current NISQ hardware?

Specifically: When applying **dynamical decoupling (DD)** or **Pauli twirling** to the idle periods and gates of a binary ALLOW/DENY authorization boundary, do we observe:

- Improved ALLOW discrimination (higher S_A)?

- Reduced DENY leakage (lower L_D)?
- Comparable or better performance vs. the unprotected baseline?

This experiment tests whether standard noise-mitigation techniques can strengthen authorization boundaries without introducing new failure modes.

2. Background & Motivation

2.1 Baseline: ARK-441

ARK-441 established a binary ALLOW/DENY boundary on `ibm_kingston` (PASS verdict):

- **S_A** (ALLOW discrimination) = 0.9979 (raw)
- **L_D** (DENY leakage, SPAM-corrected) = 0.0021
- **Δ_B** (boundary margin) = 0.9779

This was an **unprotected** baseline — no noise mitigation applied.

2.2 Noise on NISQ Hardware

IBM Quantum Heron r2 backends exhibit:

- **Decoherence** (T_1 , T_2) during idle periods
- **Gate errors** (depolarizing-like noise on single-qubit and two-qubit gates)
- **Readout (SPAM) errors** (mitigated via baseline jobs)

2.3 Noise-Mitigation Techniques

Two standard techniques:

1. **Dynamical Decoupling (DD)**: Insert sequences of π -pulses (X gates) during idle periods to average out low-frequency noise and extend coherence.
2. **Pauli Twirling**: Randomize single-qubit Pauli gates before/after operations to convert coherent errors into stochastic (depolarizing-like) noise, which is easier to characterize and correct.

2.4 Open Question

Does applying DD or Pauli twirling to an authorization boundary:

- Improve fidelity (higher S_A , lower L_D)?
- Introduce new failure modes (e.g., DD pulse errors, twirling overhead)?
- Remain stable under SPAM correction?

ARK-447 answers this by testing all three configurations on the same backend, same qubits, same protocol.

3. Experimental Design

3.1 Boundary Structure (Same as ARK-441)

Binary ALLOW/DENY authorization boundary:

- **Authorizer qubit (Q_A)**: Prepared in $|0\rangle$ (DENY) or $|1\rangle$ (ALLOW)
- **Payload qubit (Q_P)**: Target of the controlled operation
- **Boundary gate**: CNOT($Q_A \rightarrow Q_P$) with $P(Q_P) = |+\rangle$
- **Measurement**: Z-basis on both qubits

Expected outcomes (ideal):

- **ALLOW path:** $Q_A=|1\rangle \rightarrow \text{CNOT flips } P \rightarrow \text{measure } P \text{ in } |-\rangle \rightarrow \text{outcome "1"} (S_A \approx 1.0)$
- **DENY path:** $Q_A=|0\rangle \rightarrow \text{CNOT does nothing} \rightarrow \text{measure } P \text{ in } |+\rangle \rightarrow \text{outcome "0"} (L_D \approx 0.0)$

3.2 Three Configurations

Configuration 1: Baseline (unprotected)

- Same circuit as ARK-441
- No noise mitigation applied
- Acts as the control group

Configuration 2: Dynamical Decoupling (DD)

- Apply DD sequences (e.g., XY4, CPMG) during idle periods on both Q_A and Q_P
- DD sequences inserted automatically via Qiskit's `PadDynamicalDecoupling` pass
- Uses X gates (π -pulses) to refocus phase errors

Configuration 3: Pauli Twirling

- Randomize Pauli gates $\{I, X, Y, Z\}$ before and after the CNOT operation
- Twirling applied to both Q_A and Q_P
- Random Pauli pairs chosen per shot to average over all error channels
- Circuit structure: `[random Pauli] → CNOT → [compensating Pauli] → measure`

3.3 Shots & Arms

6 arms total (2 boundary conditions $\times$ 3 configurations):

1. **arm1:** ALLOW + Baseline
2. **arm2:** DENY + Baseline
3. **arm3:** ALLOW + DD
4. **arm4:** DENY + DD
5. **arm5:** ALLOW + Pauli Twirling
6. **arm6:** DENY + Pauli Twirling

Shots per arm: 8192 (same as ARK-441/445/445b)

3.4 SPAM Baseline (Gating Condition)

Two SPAM circuits (same as all ARK experiments):

- **SPAM_A:** Prepare $|1\rangle$ on Q_A , measure in Z
- **SPAM_P:** Prepare $|+\rangle$ on Q_P , measure in Z

Gate: Both must show SPAM error ≤ 0.02 to proceed.

4. Quantitative Success Criteria

4.1 Primary Metrics (per configuration)

ALLOW discrimination:

- $S_A = P(\text{outcome}=1 \mid \text{ALLOW path, given configuration})$
- **Threshold:** $S_A \geq 0.90$ (same as ARK-441/445/445b)

DENY leakage (SPAM-corrected):

- $L_{D_raw} = P(\text{outcome}=1 \mid \text{DENY path, given configuration})$

- **L_D_corrected** = $\max(0, L_D_raw - SPAM_P)$
- **Threshold:** $L_D_corrected \leq 0.02$ (strict)

Boundary margin:

- **Δ_B** = $S_A - L_D_corrected - 0.20$
- **Threshold:** $\Delta_B \geq 0.00$ (must be non-negative)

4.2 Pass/Fail Per Configuration

Each configuration (Baseline, DD, Pauli Twirling) is evaluated independently:

- **PASS:** All three criteria met ($S_A \geq 0.90$, $L_D \leq 0.02$, $\Delta_B \geq 0.00$)
- **FAIL:** Any criterion violated

4.3 Comparative Analysis

Primary comparison: DD vs. Baseline, Twirling vs. Baseline

- **Improvement:** S_A increases OR L_D decreases (with statistical significance)
- **No change:** Metrics within noise/uncertainty
- **Degradation:** S_A decreases OR L_D increases

Secondary comparison: DD vs. Pauli Twirling

- Which technique performs better on this specific backend/qubit pair?

4.4 Overall Verdict

PASS (strong): All three configurations pass individual criteria; at least one mitigation technique shows improvement over baseline.

PASS (weak): All three configurations pass individual criteria; no significant improvement observed (mitigation techniques add no harm, no benefit).

MIXED: Baseline passes; one or both mitigation techniques fail (introduces new failure modes).

FAIL: Baseline fails (boundary itself unstable on this backend/qubit pair).

5. Interpretation Boundaries

5.1 What This Experiment Tests

- ✓ Noise-mitigation impact on a binary ALLOW/DENY authorization boundary
- ✓ DD and Pauli twirling performance on IBM Quantum Heron r2 hardware
- ✓ Comparative fidelity across three configurations (same backend, same qubits, same day)
- ✓ SPAM-corrected metrics for honest leakage assessment

5.2 What This Experiment Does NOT Test

- ✗ **Generalization:** Results apply to the specific backend, qubits, and calibration used. No claim that DD/twirling will improve all boundaries on all hardware.
- ✗ **Cryptographic security:** This is a metrology experiment, not a cryptographic protocol validation.
- ✗ **Error correction:** DD and twirling are noise mitigation (not correction); no QEC codes used.
- ✗ **Multi-round or complex boundaries:** Single-round, binary boundary only.
- ✗ **Optimal DD sequences:** Uses standard DD (e.g., XY4); not optimized for this specific backend.

5.3 Known Limitations

- **DD overhead:** DD pulses add gate count, which may introduce errors if pulse fidelity is low.
- **Twirling randomness:** Pauli twirling requires many shots to average over randomness; 8192 shots may be insufficient for strong statistical separation if improvement is small.
- **Idle time dependency:** DD benefit depends on T1/T2 and the duration of idle periods; short circuits may see minimal improvement.

6. Protocol: Field 27 (Preregistration-First)

6.1 Sequence (Hard Ordering)

1. **LOCK commit** — This preregistration document, all code, and MANIFEST.txt committed to `executionproof-testbeds` on branch `execute/ark-447` with tag `ark-447-v1.0-lock` BEFORE any hardware job.
2. **Qubit selection** — Run `ark_447_select_qubits.py` to choose Q_A and Q_P based on lowest readout error sum, connected pair.
3. **SPAM baseline job** — Run `ark_447_spam_job.py` (SPAM_A and SPAM_P circuits, 8192 shots each).
Gate: Proceed only if both $\text{SPAM} \leq 0.02$.
4. **Principal job submission** — Run `ark_447_submit_ibm.py` (6 circuits $\times$ 8192 shots).
5. **Retrieve results** — Run `ark_447_retrieve.py` after job completes.
6. **Analysis** — Run `ark_447_analysis.py` to compute S_A , L_D , Δ_B for each configuration, apply SPAM correction, determine verdict.
7. **Documentation** — Generate `RESULTS.md`, `RUN_LOG.md`, and `proofrecord.json`.
8. **Tag & publish** — Tag `ark-447-v1.0`, push to GitHub, open PR.

6.2 MANIFEST Integrity

All 6 code files (`ark_447_select_qubits.py`, `ark_447_circuits.py`, `ark_447_spam_job.py`, `ark_447_submit_ibm.py`, `ark_447_retrieve.py`, `ark_447_analysis.py`) will have their SHA-256 hashes recorded in `MANIFEST.txt` and committed in the LOCK commit. Any post-LOCK modification to these files invalidates the preregistration.

6.3 No Rescue

If the experiment fails (SPAM gate fails, or any configuration fails its criteria), the result is published as-is with honest FAIL verdict. No re-running, no parameter tuning, no rescue attempts.

7. Expected Timeline

- **LOCK commit:** 2026-07-17 (today)
- **Qubit selection + SPAM gate:** ~5 minutes
- **Principal job:** ~25-30 seconds (6 circuits $\times$ 8192 shots, estimated queue time depends on backend load)
- **Retrieval + analysis:** ~2 minutes
- **Documentation + tag:** ~5 minutes
- **Total:** ~15-20 minutes end-to-end (excluding queue wait)

8. Relation to Prior Work

ARK-441 (binary ALLOW/DENY baseline, unprotected) → **PASS** ($S_A=0.9979$, $L_D=0.0021$, $\Delta_B=0.9779$)

ARK-447 tests whether DD or Pauli twirling can **improve** on ARK-441's baseline or maintain fidelity while adding robustness.

If ARK-447 shows improvement, it suggests noise mitigation can strengthen authorization boundaries.

If ARK-447 shows no change, it suggests the baseline is already near the hardware limit.

If ARK-447 shows degradation, it suggests mitigation techniques introduce overhead that outweighs their benefit on this backend.

9. Authorship & Provenance

- **Principal Investigator:** Derek Hone, Remnant Fieldworks Inc.
 - **Organization:** Remnant Fieldworks Inc. (ExecutionProof product track)
 - **Repository:** `derekhone/executionproof-testbeds`
 - **Preregistration commit:** To be tagged `ark-447-v1.0-lock` before hardware execution
 - **Backend provider:** IBM Quantum (open-access Heron r2 backends)
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10. Disclosure

No conflicts of interest. This experiment is part of the ExecutionProof authorization-boundary R&D program. Results will be published regardless of outcome (PASS/FAIL/MIXED). No financial incentive to suppress negative results.

Reproducibility: Full code, circuits, raw counts, and calibration snapshots will be published in the `executionproof-testbeds` repository. Any researcher with IBM Quantum access can attempt to reproduce this experiment on the same or different backends.

This preregistration is locked before hardware execution. Any changes to success criteria, circuit design, or analysis methodology after hardware jobs are submitted invalidate the preregistration protocol.